

Polynomials – 2

Integer coefficients, rational roots and the shape of the
coefficients

Homework

8 сентября 2026 года

Theoretical minimum. The divisibility $a - b \mid f(a) - f(b)$; the rational root theorem; Gauss's lemma and factorisation over $\mathbb{Z}$ versus over $\mathbb{Q}$; conjugate roots of an irreducible polynomial; recovering coefficients from values; the sum of the squares of the coefficients as an integral over the unit circle.

1. (Putnam 1976) Let r be a root of $P(x) = x^3 + ax^2 + bx - 1$ and let $r + 1$ be a root of $Q(y) = y^3 + cy^2 + dy + 1$, where a, b, c, d are integers and P is irreducible over $\mathbb{Q}$. Express another root s of P as a function of r not involving a, b, c, d .
2. (Putnam 1967) Consider quadratic polynomials $ax^2 - bx + c$ with integer coefficients having two distinct zeros in the open interval $(0, 1)$. Exhibit, with proof, the least positive integer value of a for which such a polynomial exists.
3. (Putnam 1985) Let k be the smallest positive integer with the following property: there are distinct integers $m_1, \dots, m_5$ such that $p(x) = (x - m_1)(x - m_2)(x - m_3)(x - m_4)(x - m_5)$ has exactly k non-zero coefficients. Find, with proof, a set of integers $m_1, \dots, m_5$ for which this minimum is achieved.
4. (Putnam 1985) For a real polynomial $p(x) = a_0 + a_1x + \dots + a_mx^m$ put $\Gamma(p) = \sum_{k=0}^m a_k^2$, and let $f(x) = 3x^2 + 7x + 2$. Find, with proof, a polynomial g with real coefficients such that $g(0) = 1$ and $\Gamma((f(x))^n) = \Gamma((g(x))^n)$ for every integer $n \geq 1$.

5. (Putnam 2017) Let $Q_0(x) = 1$, $Q_1(x) = x$ and $Q_n(x) = \frac{(Q_{n-1}(x))^2 - 1}{Q_{n-2}(x)}$ for $n \geq 2$. Show that $Q_n(x)$ is always a polynomial with integer coefficients.
6. (Putnam 2022) Suppose that $P(x) = a_1x + a_2x^2 + \cdots + a_nx^n$ has integer coefficients with a_1 odd, and that $e^{P(x)} = b_0 + b_1x + b_2x^2 + \cdots$ for all x . Prove that b_k is non-zero for every $k \geq 0$.
7. (Putnam 2023) Let n be an even positive integer and let p be a monic real polynomial of degree $2n$ with $p(1/k) = k^2$ for all integers k with $1 \leq |k| \leq n$. Find all other real numbers x for which $p(1/x) = x^2$.
8. (Putnam 2008) Let $n \geq 3$ and let f and g be real polynomials such that the points $(f(1), g(1)), \dots, (f(n), g(n))$ are the vertices of a regular n -gon in counterclockwise order. Prove that at least one of f and g has degree at least $n - 1$.
9. (Putnam 2010) Find all pairs of real polynomials $p(x)$ and $q(x)$ for which $p(x)q(x+1) - p(x+1)q(x) = 1$.
10. (Putnam 2022) Let $n \geq 2$ be an integer. Over all real polynomials $p(x)$ of degree n , what is the largest possible number of negative coefficients of $(p(x))^2$?